

Sagar Mandiya

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Education

BOSTON UNIVERSITY

Master of Science, Computer Science

09/2022 - 06/2024 (expected)

- Relevant Courses – Artificial Intelligence, Foundation Machine Learning, Computer Language Theory, Web Mining and Analytics.

ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Bachelor of Technology, Computer Science and Engineering

07/2017 - 06/2021

- Relevant Courses – Object Oriented Programming & Methodology, Analysis Design of Algorithm, Pattern Recognition, Machine Learning, Theory of Computation, Internet of Things, Cloud Computing, Image Processing & Computer Vision.

Experience

BOSTON UNIVERSITY

MA, USA

Graduate Research Assistant

03/23 - Present

- Conduct research on music generation using generative AI.
- Research cause and effect of generative ai in today's market with respect to content creation as a whole.

Graduate Teaching Assistant

Jan. 2023 - APR. 2023

- Conduct office hours for the class of BA820 - Unsupervised & Unstructured Machine Learning.
- Help Professor in developing learning activities.

TENSORGO TECHNOLOGIES PRIVATE LIMITED

Hyderabad, India

Computer Vision Engineer

05/22 - 08/22

- Worked extensively on virtual perimeter overlay; computer vision project involving image segmentation and object detection.
- Developed an in-depth understanding of model tuning for optimal performance on image related problems.

Associate Computer Vision Engineer

09/21 - 05/22

- Designed project pipeline for a complex deep learning based project from scratch and implemented it effectively.
- Constructively worked with team members as a project manager and developer and increased product efficiency by 20%.
- Effectively communicated with team members and stakeholders to resolve problems at an early stage.
- Promoted within an year for excellent performance.

Publication

Neural networks with rhythmic flow* :

(2022)

- Working on speech-text synthesis using Nvidia NeMo ASR.

Deep Fake Detection: Survey of Facial Manipulation Detection Solutions: :

(2021)

Published in International Research Journal of Engineering and Technology (IRJET) (arxiv.org/abs/2106.12605)

- Studied the advances in the **Facial Manipulation** field, and trained state-of-the-art networks for **performance analysis**.

* On-Going Publication

Academic Projects

Answer Verification System - TensorGo Technologies Private Limited

-Team size: 3

- Working as a project manager on developing a **deep learning** based model for the use case of interviews, to verify the response given by interviewees and assign them a relevance score.

Automatic Speech Recognition and Voice Biometrics - TensorGo Technologies Private Limited -Team size: 2

- Developed a **deep learning** based model to transcribe the speech and recognize the speaker based on their pre saved embeddings, using Nvidia NeMo

Face Mask and Social Distancing Detection- TensorGo Technologies Private Limited -Team size: 1

- Designed and integrated the **operation cycle, and face detection caffe model** on stand-alone unit meant to detect if the social distancing and face mask guidelines are being followed at public places.

Deep Fake Detection - Prof. Rashid Sheikh & Prof. Vandana Kate -Team size: 3

- Executed the **data collection, pre-processing, training, fine-tuning and analyzed** the XceptionNet, VGG-19 and Mesonet models for better performance on the use case of detecting fake images and videos circulating the internet.

Movie Recommendation System -Team size: 1

- Developed a **machine learning** based **recommendation system** to suggest movies based on factors like genre and IMDB ratings using cosine similarity and other techniques.

Grocery Product detection -Team size: 1

- Developed a **Convolutional Neural Network** based model using **TensorFlow Object Detection** library to list total types and brands of items present on the grocery shelf and record the results on a file.

Sarcasm Detection in News Headlines -Team size: 1

- Developed a **bi-directional LSTM** based model capable of detecting sarcasm from the news headlines.

Portfolio Website and Blog -Team size: 1

- Developed a portfolio website cum blog from scratch using HTML5, CSS3, and JAVASCRIPT, which can be visited at <https://www.sagarmandiya.com/>

Technical Skills

- **Programming Languages:** C, C++, Python, R, HTML, CSS, Javascript, PHP, SQL.
- **Libraries:** Numpy, Pandas, OpenCV, Imutils, Matplotlib, Scikit-learn.
- **Frameworks:** Nvidia CUDA, TensorRT, PyTorch, TensorFlow.
- **Operating Systems:** Ubuntu, Kali linux, MacOS, Windows.
- **Additional Skills:** WordPress, Bootstrap, Git.

Certifications

- **Deep Learning Specialization** by deeplearning.ai.
- **Machine Learning** by Stanford.
- **Convolutional Neural Networks in TensorFlow** by deeplearning.ai.
- **Introduction to TensorFlow for AI, ML, and DL** by deeplearning.ai.
- **Databases and SQL for Data Science** by IBM.
- **The Bits and Bytes of Computer Networking** by Google.
- **Python Data Structures** by University of Michigan.

* Click on course title to access the certification

Activities

- Participated in the **Red Cross Society** blood donation camp under **National Service Scheme (NSS)**, 2020.
- Organized a **charity play-zone** named "Khel-Adda" in annual university cultural festival to raise funds which were donated to the families of martyrs of 2019 Pulwama Attack, 2019.
- **Armageddon: Annual Technical Festival of AITR, Indore**, 2019.
 - Integral part of the **organizing team** of national level Tech-Fest.
 - Organized **Technical Quiz** in the Tech-Fest to encourage the students to dig deep into the field of computer science.
- Qualified for the second round of International Coding Competition organized by Codechef, 2019.